Lecture 21: Three-Dimensional Coordinate Systems

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

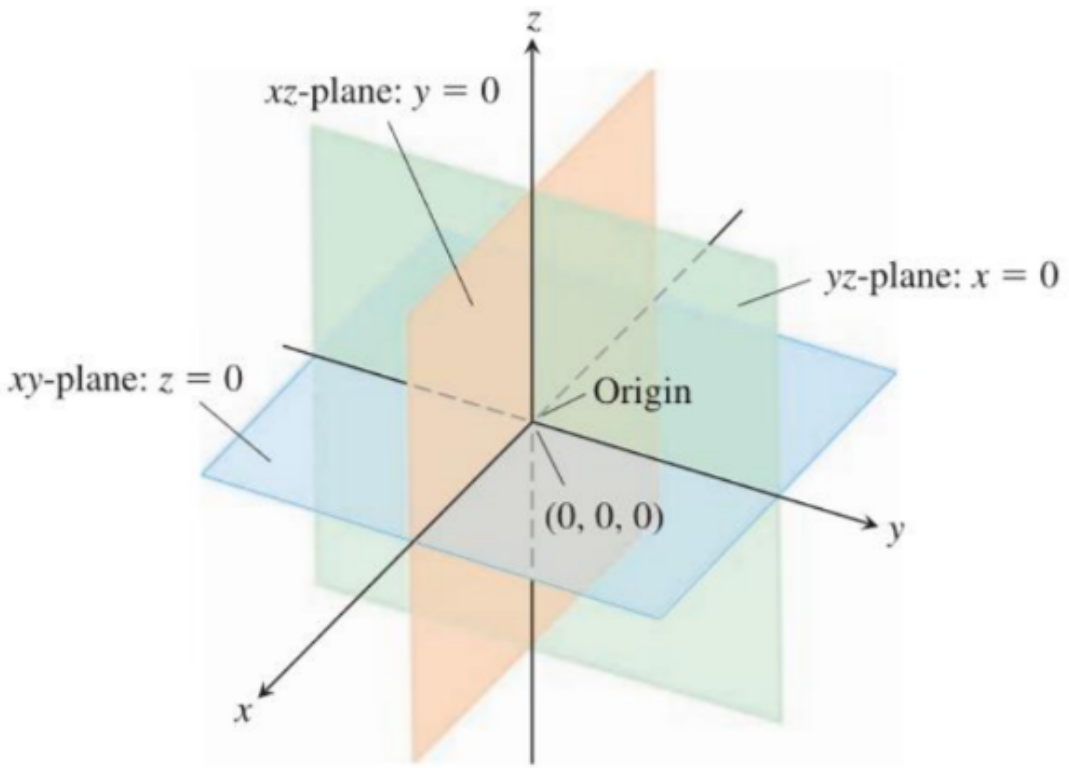
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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define and identify the three-dimensional Cartesian coordinate system.
- ☐ Locate points in space using ordered triples (x, y, z).
- ☐ Describe the coordinate axes, coordinate planes, and the concept of octants.
- ☐ Determine the distance between two points in space.
- ☐ Recognize the equation of a sphere given its center and radius.

3-Dimensional Coordinates System



3-Dimensional Coordinates

Example: Give the geometric interpretation of the following equations and inequalities.

(a) $z \geq 0$

(b) $x = -3$

(c) $z = 0, x \leq 0, y \geq 0$

3-Dimensional Coordinates

(d) $x \geq 0, y \geq 0, z \geq 0$

(e) $-1 \leq y \leq 1$

(f) $y = -2, z = 2$

3-Dimensional Coordinates

Example: What points (x, y, z) satisfy the equations

$$x^2 + y^2 = 4 \quad \text{and} \quad z = 3?$$

Distance Between Points in Space

The Distance Between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance Between Points in Space

Example: Find the distance between $P_1(2, 1, 5)$ and $P_2(-2, 3, 0)$.

Spheres

The collection of points with the same distance from a particular point forms a sphere.

Standard Equation for the Sphere of Radius a and Center (x_0, y_0, z_0)

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

Spheres

Example: Find the center and radius of the sphere

$$x^2 + y^2 + z^2 + 3x - 4z + 1 = 0.$$

Spheres

Example: Give the geometric interpretation of the following equations and inequalities.

(a) $x^2 + y^2 + z^2 < 4$

(b) $x^2 + y^2 + z^2 \leq 4$

(c) $x^2 + y^2 + z^2 > 4$

(d) $x^2 + y^2 + z^2 = 4,\ z \leq 0$